PROBLEM STATEMENT CODE: PS000109

WorkCheck Kachchh (કચ્છ શ્રમિક સુરક્ષા)

"Finding exposed unorganised workers through the **public health system**, triggering preventive screenings via **low-cost environmental sensing**, and following every referral to completion into an **ABHA-linked worker-owned health pass**."

TEAM ID	TRACK CATEGORY	INSTITUTION / ORGANIZATION	TEAM LEADER
TM005024	College Track (Engineering)	LDRP Institute of Technology & Research, Gandhinagar, Gujarat	Jay Patel (Tean

CHALLENGE DEFINITION

Problem Statement & Real-World Context

Focus: Roo

The Ground Challenge & Evidence

Challenge: Kachchh has over **1.8 lakh unorganised workers** across 10 talukas (salt rakers, quarry drillers, port loaders, cement yard workers) who never appear on formal factory registers. Current rules (**Gujarat OSH Rule 9 / Central Rule 5**) mandate check-ups only by employer type and age (**45+ / 40+**), yet dust and heat harm through cumulative exposure regardless of age. Inspection coverage is merely **15% (2021 data)**, and occupational disease goes unrecorded (only **562 cases** nationally in 2011–16 vs. **182 silicosis cases & 80 deaths** found by a single Gujarat NGO, PTRC). Laborers avoid reporting symptoms due to fear of wage loss or termination.

EMPIRICAL EVIDENCE (FIELD & REGULATORY DATA)

- **87.5% of salt-pan workers** exceed heat stress limits with 2.1× kidney impairment risk (Venugopal et al.).
- **26% of industrial worker samples** exceed the silica PEL (50 µg/m³).
- **Kandla port & Vatva:** Kandla had 2,067 daytime hours >27.5°C WBGT in 2025; Vatva exceeded PM10 limits on 83% of days.

Target Beneficiaries & Flaws

Beneficiaries (Primary Field Cohorts):

- **1) Agariya Salt Rakers (Little Rann & coastal pans):** Expose radiant heat & white salt-crust reflection.
- **2) Stone Crusher & Quarry Drillers:** Exposed to high respira crystalline silica dust.
- **3) Dock & Port Loaders (Kandla & Mundra):** Heavy outdoo tropical marine WBGT.
- **4) Mobile Health Clinicians:** Doctors aboard BOCW Dhanvan mobile vans.

EXISTING MARKET & SYSTEMIC GAPS

- **Wearable Patches (Kenzen, SlateSafety):** Cost ~₹4,100/worke for informal labor.
- **Industrial Monitors (TSI DustTrak):** Cost ₹5.5 Lakhs per uni referral links.
- **Employer Software (OccuCare, OHCTECH):** Caters strictly bypassing informal labor.

SOLUTION APPROACH •

Proposed Solution & Innovation Novelty

Ethos: F

HIGH-LEVEL EXECUTIVE SUMMARY • DIGITAL-FIRST, SENSOR-OPTIONAL SURVEILLANCE

WorkCheck Kachchh adds a **1-minute exposure check** to Ayushman Arogya Mandir visits, audits industrial clusters via **₹2,015 solar ambient IoT nodes** (+ optical dust), triages workers through **3 clinician-approved routing rules**, and publishes statutory records directly into the worker's **ABHA Health Locker**

Core Novelty

- **Worker-Owned Health Pass:** Records follow the laborer via 12-digit e-Shram UAN and ABDM ABHA ID across employers.
- **Past Exposure Awareness:** Evaluates cumulative occupational years (e.g., 20 years at a crusher), not just today's sensor reading.
- **100% DPDP Act 2023 Compliance:** OTP consent-gated records; district administration sees strictly anonymous statistical heatmaps.

Technical Moat

- **ISO 7243 WBGT Edge Engine:** 150mm matte-black copper globe avoids underestimation of true solar heat under wind.
- **Coastal Humidity Correction:** Aerosol correction prevents humidity fogging in optical PM sensors ($R^2 >$ 0.88 vs OPC-N3).
- **Native HL7 FHIR R4 Bundle Generator:** 1-click publishing of DiagnosticReports to ABDM Health Locker.

"Seva First" Va

- **Direct Grassroots Welfare:** Ze Operating cost is < ₹7.30 / work than US alternatives).
- **100% Fundable via Cess:** Bac Construction Welfare Cess & D (PMKKKY).
- **Trilingual Access:** Native inte (ગુજરાતી), Hindi, and English.

LIVE SANDBOX & DEMO

Proposed Solution Explanation / Demonstration Video & Live Evaluator Sandbox:
https://jaypatel29042008-glitch.github.io/WorkCheck-Kachchh-SSIP/

SYSTEM ARCHITECTURE •

Technical Architecture || Methodology

① SENSING & TELEMETRY
Edge Hardware & Microclimate (Teal)

Ambient IoT Site Node (₹2,015 BOM)
Solar-powered ESP32 node at salt pans, quarries, and port docks.

150mm Matte-Black Copper Globe
Measures true solar radiant heat (Tg) without wind bias (R^2 = 0.92 vs 0.65 for 40mm ball).

Multi-Hazard Sensor Array
PMS5003 laser particle counter + Bosch BME280 (Ta, RH) + GUVA-S12SD UV sensor.

WS2812B RGB Industrial Flag Lamp
Visual work-rest schedule signaling on site (GREEN / YELLOW / ORANGE / RED).

Zero-Sensor Graceful Fallback
Core health screening & referral tracking operate 100% without hardware.

② RULES & COMPUTATION
SQL DB & 3 Clinical Rules (Orange)

ISO 7243 WBGT Calculator
$WBGT_{outdoor} = 0.7\,T_{nw} + 0.2\,T_g + 0.1\,T_a$ (ACGIH heavy labor thresholds).

Coastal Aerosol Correction
$PM_{corr} = PM_{raw} / [\,1 + 0.25 \cdot (RH^2 / (1 - RH))\,]$ (prevents humidity fogging).

10-Table ANSI SQL Database
30-day raw high-frequency telemetry buffer and permanent longitudinal shift rollups.

3 Clinical Routing Rules
1) Statutory (Rule 9/5), 2) Cumulative Quartz, 3) Symptoms (Dark urine, cough).

Clinician Ownership Safeguard
Software only flags occupational risk; doctors aboard mobile vans make all diagnoses.

③ DELIVERY & GOVERNA
QR Pass, Mobile Van & DDO (Eme

Worker QR Health Pass
Bilingual card encoded with 12-dig
follows worker across employers.

BOCW Dhanvantari Arogya R
Mobile van clinical station for spir
serum creatinine, and ILO X-ray.

1-Click ABHA FHIR Publisher
Deposits HL7 FHIR R4 Diagnosti
into worker's ABHA locker.

Form 5 Annual Fitness Certific
Automated statutory fitness certifi
Gujarat OSH Rules 2025.

DDO Kachchh Macro View
Anonymized Leaflet GIS taluka he
CSV exports (BOCW / NPCCHH)

ENGINEERING STACK •

Technology Stack & Feasibility

Component

A. EMBEDDED IOT & HARDWARE

TRL-5 • VALIDATED

- **Controller & Edge Buffer:** ESP32-WROOM-32 (240MHz dual-core, WiFi/BLE, offline flash ring buffer).
- **Environmental Sensors:** Bosch BME280 (Ta, RH), Plantower PMS5003 laser PM, GUVA-S12SD UV.
- **Radiant Heat Globe:** 150mm matte-black copper float globe + 10k NTC thermistor.
- **Solar Power & Unit BOM:** 5V 2A Solar + 18650 Li-ion + IP65 box. Unit BOM: ₹2,015 (bulk).

B. BACKEND & RULES ENGINE

TR

- **Async REST Framework:** Python 3.10+, FastAPI async REST OpenAPI/Swagger.
- **Database Architecture:** 10-table ANSI SQL schema (SQLite/P raw buffer.
- **Scientific Evaluators:** ISO 7243 ACGIH work-rest evaluator + correction.
- **Automated Test Suite:** 21/21 automated integration tests passir roles.

C. FRONTEND & CLINICAL UI

TRL-6 • LIVE PWA

- **Progressive Web App:** Mobile-first responsive PWA operating offline or connected via Tailwind CSS.
- **GIS & Telemetry Charts:** Leaflet.js interactive taluka GIS heatmap + Chart.js live sensor graphs.
- **Trilingual Localization:** Native Gujarati (ગુજરાતી), Hindi, and English interfaces across 4 personas.
- **Clinical Van Station:** Spirometry (FEV1/FVC), serum creatinine, ILO X-ray & Form 5 generator.

D. STANDARDS & PRIVACY

TR

- **ABDM Interoperability:** ABDM HIP/HIU sandbox integration DiagnosticReport bundles.
- **Portable National ID:** 12-digit e-Shram UAN QR health pass f portability.
- **Empirical Data Catalog:** 10 public datasets cataloged (ERA5, OSHA, PLFS).
- **DPDP Act 2023 Privacy:** 100% compliant; OTP worker conser DDO heatmaps.

IMPACT ASSESSMENT • SIH SECTION 05

Socio-Economic Impact & Viability

Qua

1. CLINICAL IMPACT

>90% Follow-Up

2.1× Kidney Risk Addressed

Prevents AKI & CKDu

Protects salt rakers (87.5% heat stress, 2.1× kidney risk) via hydration & urine checks.

Early Silicosis Capture

Detects cumulative silica exposure in quarry drillers years before irreversible lung fibrosis.

Closes Referral Leakage

Overcomes the 15% inspection gap by tracking every referred case to completion (>90% follow-up).

2. FRUGAL ECONOMICS

₹7.30 / Month

560× Cheaper Than US Patches

560× Cost Advantage

Operates at ₹7.30 / worker / month vs ₹4,100 / month for commercial US patches (Kenzen).

₹2,015 Site Node

Replaces ₹5.5 Lakh TSI DustTrak monitors with calibrated, humidity-corrected ambient nodes.

District Pilot Budget

Covers 50 worksites & 1,000 workers for ₹2,97,750 Year 1 (ongoing ₹88 / worker / year).

3. PRODUCTIVITY

~358 Hrs/Yr

~20% Severe Heat Hours Saved

Cool-Shift (6am–3pm)

Moving 9am–6pm shifts to 6am–3pm avoids ~20% severe heat (~358 hot hours saved/yr).

Objective Rest Flags

Automated ISO 7243 RGB flag lamps eliminate subjective employer work-rest disputes.

Mandatory Amenities

Audits potable drinking water and shaded rest shelters at every site node.

4. GO

₹0 W

100% BOCW

Zero Cost to W

Laborers pay ₹0; strictly protected sold.

100% Fundabl

Backed by Gujar Construction We Mineral Foundat

Statutory Enfo

Operationalizes Central Rule 5, a Plans.

EXECUTION TIMELINE • SIH SECTION 06

Implementation Roadmap & Milestones

From

STAGE 1: LAB VALIDATION

Oct – Nov 2026 • TRL-5
STATUS: COMPLETED

Empirical Datasets
Analyzed ERA5, CAMS, CPCB & OSHA empirical datasets across 5 Kachchh sites.

Database & Engine
Engineered 10-table ANSI SQL DB & ISO 7243 WBGT rules engine.

4-Persona Web App
Built interactive dashboard with 1-click ABHA FHIR R4 publishing.

Verified & Live
Verified 21/21 integration tests; deployed live on GitHub Pages.

STAGE 2: PILOT FIELD TRIAL

Nov – Dec 2026 • TRL-6
STATUS: PLANNED

Field Site Nodes
Deploy 2–3 calibrated solar nodes at Rapar (salt pan) & Abdasa (quarry).

Reference Validation
Validate black-globe & PM telemetry against reference instruments.

PHC / CHC Pilot
Pilot 1-minute occupational exposure check at 1 PHC / CHC.

Clinical Rule Review
Clinical rule review with CDHO Kachchh and NIOH Ahmedabad.

STAGE 3: VAN INTEGRATION

Jan – Mar 2027 • TRL-7
STATUS: PLANNED

Mobile Van Fleet
Direct integration with Gujarat BOCW Dhanvantari Arogya Rath mobile vans.

50-Site Rollout
Deploy 50 solar nodes covering 1,000 unorganised laborers.

QR Health Passes
Issue bilingual PVC QR Health Passes encoded with e-Shram UAN.

Live ABHA Deposits
Live deposit of spirometry & creatinine records into ABHA Lockers.

STAGE 4:

Apr 2027
STAT

10-Taluka Scal
Scale surveillanc
Talukas (10,000

ESIC Referral
Automated refer
Dispensaries (Bl

Heat Micro-In
Connect site tele
parametric heat

District Policy
Present anonymi
Kachchh for dist

SQUAD CAPABILITIES

Multi-Disciplinary Team & Mentorship

TL

Jay Patel

Team Leader & Architect

- **System Architecture:** End-to-end 3-layer surveillance platform design.
- **Database & Rules:** 10-Table ANSI SQL DB & ISO 7243 WBGT engine.
- **ABDM FHIR R4:** 1-Click ABHA DiagnosticReport publishing service.
- **Test Harness:** Automated suite (21/21 integration tests passing).

M2

Hardware & IoT Lead

Embedded & Calibration

- **ESP32 Firmware:** Dual-core C++ firmware & Wokwi circuit simulation.
- **Globe Calibration:** 150mm matte-black globe radiant heat curve.
- **Sensor Wiring:** PMS5003 dust, BME280 & GUVA-S12SD UV array.
- **BOM Optimization:** Solar + Li-ion power & ₹2,015 bulk BOM design.

M3

Public Health Lead

Field Ops & Localization

- **e-Shram UAN Flow:** 12-Digit QR health pass workflow across employers.
- **Mobile Van Liaison:** BOCW Dhanvantari Arogya Rath route scheduling.
- **Trilingual UI:** Native Gujarati (ગુજરાતી), Hindi & English UI.
- **Frontline Triage:** ASHA/ANM 1-minute exposure check workflow.

Comp

Hygiene

- **Gujarat OSH** Form 5 fitness
- **Legal Mappi** 41C & Centra
- **DPDP Act 20** audit & k-ano
- **Public Datase** cataloged & m

Faculty Mentor: Prof. / Dr. Faculty Mentor

Department of Computer Engineering / IT, LDRP-ITR, Gandhinagar

Ready for Ju